

T4 Operational Concept Disambiguation Synthesis

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This work derives from and formalizes commitments across the CKS theory trilogy: “The Instinct/Reasoning Separation Outside the Model” (Li, April 2026), “Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems” (Li, April 2026), and “Inter-Self Coordination via Shared Substrate / Full Aspect Integration” (Li, April 2026).

Abstract

This note synthesizes the five operational concept disambiguation notes of the T4 subseries (T4.01–T4.05), which addressed provenance/record, the three governance rights, scope, accountability and traceability, and governance record completeness. The synthesis identifies two findings that hold across all five notes. First, a pattern finding: the operational governance concepts of the trilogy carry a consistent meaning throughout all three papers; the differences across papers are differences in scope, not differences in concept. Second, a formalization finding: Paper 3 formalizes what Papers 1 and 2 implied; its explicit specifications — the seventeen-category governance record taxonomy, the four completeness standards, and the four-link authorization chain — are operational precision over commitments the earlier papers already made, not the introduction of new concepts. This formalization finding is the T4 parallel to T3's architectural stability finding. One exception is noted: sharing scope (T4.03) is a Paper 3 technical term with no precise equivalent in Papers 1 or 2, and is the T4 subseries' one genuinely new-at-Paper-3 concept. The note concludes by transitioning to T5, which addresses population and scale concepts that are structurally different from T4 concepts: they are primarily Paper 3 Claim 6 terms, and their disambiguation is within-Paper-3 clarification rather than cross-paper reconciliation.

1. Purpose and scope of this synthesis

T4.01 through T4.05 formed a focused subseries on operational concept disambiguation — specifically, on how the trilogy's governance operation terms behave across papers. These are not the structural or architectural concepts addressed in T1–T3 (substrate, cell, aspect,

Self, conflict-handling, evolution, governance perimeter, authority). They are the operational vocabulary practitioners use when managing a governed substrate: what a governance record is, what rights the governing humans hold, what scope a governance action covers, how accountability is maintained, and what it means for a governance record to be complete.

The disambiguation task for operational concepts has a different character than for structural concepts. Structural concepts extend to larger scopes and new roles as the trilogy develops; their disambiguation traces how the same architectural primitive appears at successively larger rungs. Operational concepts, by contrast, do not principally change scope — they deepen in specification. A governance record is a governance record at Paper 1 cell scope and at Paper 3 cross-organizational scope; what changes is how much the paper specifies about what must be in it. This note draws out that pattern and its one partial exception.

2. What T4.01–T4.05 established

T4.01 — Provenance/Record.

Six provenance metadata fields apply consistently throughout the trilogy. Paper 3 adds two FAI-origin provenance dimensions (D1.22 Dimensions 5 and 6) specific to cross-organizational substrate contribution, and introduces a four-link authorization chain for inter-Self contexts. The Paper 3 term “governance record” — when used at Paper 3 scope — carries the specific seventeen-category taxonomy that T4.01 identified. Beneath that taxonomy, the same six-field structure from Paper 1 remains the provenance foundation.

T4.02 — The Three Rights.

The three governance rights (inspect, modify, override) are unconditional at every scope throughout the trilogy. Paper 3 extends them: at inter-Self scope, Selves hold joint rights over the shared substrate and home rights over their home substrate simultaneously, and the exit right extends override to the relationship scope. Neither the unconditional character of the rights nor their identity as the same three rights changes at any scope.

T4.03 — Scope.

“Scope” in the trilogy carries four distinct referents: tier scope (the ladder rung at which a commitment operates), governance authority scope (the set of practitioners subject to a governance arrangement), sharing scope (Paper 3’s configured dimension governing what content a Self contributes to the shared substrate), and scale shorthand (informal usage in population-level discussion). Sharing scope is the operationally critical Paper 3 technical term; it has no precise equivalent in Papers 1 or 2.

T4.04 — Accountability/Traceability.

Traceability is a mechanism — the provenance chain that makes responsible practitioners identifiable. Accountability is a principle — that the practitioners who made governance decisions can be identified. Non-delegation protects accountability by ensuring the humans who hold governance authority cannot offload that authority without a traceable record. The four-link authorization chain at Paper 3 scope extends both mechanism and principle to cross-organizational contexts, without altering the traceability-mechanism/accountability-principle distinction.

T4.05 — Governance Record Completeness.

The existence of a governance record is not the same as its completeness. T4.05 identified four D2.36 completeness standards that define what a governance record must contain at Paper 3 scope, and named AP-21 as the anti-pattern for records that exist without meeting those standards. Papers 1 and 2 did not specify completeness standards explicitly; Paper 3's four standards formalize the completeness dimension that path retraceability in Paper 1 implicitly required.

3. The T4 disambiguation pattern

Across all five T4 notes, a consistent pattern holds.

Concept consistent. The operational governance concepts — provenance, the three rights, accountability, traceability, governance records — carry the same meaning throughout the trilogy. There is no semantic drift across papers for any of them. A practitioner using the term “provenance” or “accountability” or “the three rights” at Paper 1 scope and at Paper 3 scope is referring to the same concept; the difference is the scope at which it applies and the detail with which the paper specifies it.

Formalized at Paper 3. Paper 3 provides the most explicit formalization of all five operational concepts. Where Papers 1 and 2 established the concepts and committed to them as architectural properties, Paper 3 specifies exactly what a complete governance record contains (seventeen categories), what completeness means (four D2.36 standards), what the authorization chain looks like across organizational boundaries (four links), and what the configured dimension governing substrate contribution is called (sharing scope). These specifications are not new concepts. They are the operational precision that the cross-organizational scope makes necessary.

4. The T4 formalization claim

Paper 3 formalizes what Papers 1 and 2 implied. The most direct illustration is governance record completeness. Paper 1 commits to path retraceability — the substrate must support

full reconstruction of any decision's authorization and execution path. That commitment implies that governance records exist and that they contain enough to support retracing. Paper 3 specifies exactly what “enough” means: four completeness standards and a seventeen-category taxonomy for record content. The standards are not new requirements; they are the explicit form of the implicit requirement Paper 1 already carried.

The same structure holds for the four-link authorization chain (Paper 3's formalization of the accountability principle Paper 1 established) and for the two FAI-origin provenance dimensions (Paper 3's formalization of provenance requirements that arise when substrate contribution crosses organizational boundaries, under the same six-field structure Paper 1 introduced).

This formalization finding is the T4 parallel to T3's architectural stability finding. T3 found that the architectural commitments of the trilogy — substrate-mediator, conflict-as-first-class, human governance — hold stably as scope expands from cell to Self to inter-Self. T4 finds that the operational governance commitments hold stably as scope expands, with Paper 3 providing the explicit operational specifications that the cross-organizational context requires. Formalization is operational precision over existing commitments, not conceptual change.

5. Sharing scope as T4's distinctive contribution

One partial exception to the consistency pattern must be named. Sharing scope (T4.03) is a Paper 3 technical term — a governance-configured dimension that determines what content a Self contributes to the shared substrate and under what conditions. It has no precise equivalent in Papers 1 or 2. A Paper 2 concept of content-domain scope is adjacent but not identical; Papers 1 and 2 do not configure a dimension for cross-organizational substrate contribution because no cross-organizational substrate exists at those scopes.

Sharing scope is therefore the T4 subseries' one genuinely new-at-Paper-3 concept. The other four operational concepts (provenance/record, the three rights, accountability/traceability, governance record completeness) are consistent extensions of concepts that appear — with varying degrees of explicitness — in all three papers. Sharing scope is specific to the inter-Self substrate architecture and emerges only when that architecture is introduced in Paper 3.

This does not weaken the formalization claim for the other four concepts. It sharpens the claim by identifying exactly where Paper 3 introduces something that is not a formalization of a prior implication: the configuration dimension itself is new, though the governance principle that it must be human-authored and subject to the three rights follows directly from Paper 1 and 2 commitments.

6. Transition to T5

T5 (approximately five notes, beginning with T5.01) will address population and scale concepts: how terms such as “network,” “topology,” “population,” “collective evolution,” “propagation,” “convergence,” and “diversity” are used in Paper 3’s Claim 6, and what each means within that context.

T5 has a structurally different character from T1–T4. In T1–T4, the disambiguation task was primarily cross-paper: tracing how concepts that appear across multiple papers carry consistent meanings or consistent extensions as scope expands. Most T1–T4 concepts appear — at least implicitly — in all three papers, and the disambiguation notes could draw on all three papers as sources. T5’s population/scale concepts are primarily Paper 3 Claim 6 terms. Papers 1 and 2 do not operate at population scope; their architectures concern the cell, the Self, and the inter-Self boundary.

The T5 disambiguation task is therefore primarily within-Paper-3 clarification: establishing precisely what each population/scale term means within Paper 3, distinguishing terms that might appear synonymous (such as “convergence” and “propagation”), and identifying which concepts are scope extensions of intra-Self concepts and which are genuinely novel at population scope. Where cross-paper comparison is possible — for example, where Paper 3 population-scope concepts have roots in Paper 2 Self-scope evolution mechanisms — T5 notes will draw those connections. But the primary disambiguation work in T5 is definitional within Paper 3 rather than reconciling meanings across the trilogy.

Source papers

Li, W. (2026). *The Instinct/Reasoning Separation Outside the Model*. April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems*. April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). *Inter-Self Coordination via Shared Substrate / Full Aspect Integration*. April 2026. ORCID: 0009-0004-8065-3235.

Related notes in this series

T4.01 — Provenance/Record Disambiguation (Series T)

T4.02 — Three Rights Disambiguation (Series T)

T4.03 — Scope Disambiguation (Series T)

T4.04 — Accountability and Traceability Disambiguation (Series T)

T4.05 — Governance Record Completeness Disambiguation (Series T)

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